

BrockMartinLab2

Memorandum

The Minnesota ROTC program has contacted us in determining the equitability of their obstacle course. Their main concern is regarding that the Cadets' heights are making the obstacle course either easier or harder to complete. This memorandum will address the ROTC's concerns for the obstacle course as well as looking into other factors that are causing difficulties completing the obstacle course.

Figures 01, 02, and 03 shows the relationship that height, weight, and completion time have with each other. Addressing the main concern, it does appear that shorter Cadets have a more difficult time completing the course than taller Cadets. However, when considering the Cadets' gender the results are a lot more clearer. *Figures 04, 05, and 06* are the same scatter-plots as *Figures 01, 02, and 03* but grouped by gender. By grouping the Cadets' gender we see that a lot of female Cadets are experiencing difficulties completing the obstacle course than the male Cadets. More specifically, taller female Cadets are able to complete the course faster than shorter female Cadets can. While shorter male Cadets can complete the course at a similar pace than the taller male Cadets can. Looking at *R Code 08, 09, and 10*, the analysis of four distinct groups of Cadets based on gender and height. We do see that differences in heights are significant for both male and female Cadets. From the perspectives of the shorter Cadets, there is significant evidence that they complete the obstacle course slower than their fellow counter-parts. However, the slowest predicted difference in heights for male Cadets (12.29 seconds slower) is nearly twice as fast than the fastest predicted difference in heights for female Cadets (23.81 seconds slower) from *R Code 10*. So it does appear that taller Cadets, regardless of gender have an advantage in completing the obstacle course over the smaller Cadets.

Recommendations that the ROTC can utilize in creating an equitable obstacle course is to gather more data from the course itself. We only have the total time it took Cadets to complete the entire course, not where the Cadets' are having a particularly difficult time at. More info like type of obstacle, type of material or terrain used, and Cadet's time to complete that part of the obstacle course are crucial elements to be considered during this analysis. Following along with shorter Cadets are experiencing more difficulties in completing the course compared to their fellow Cadets. The Minnesota ROTC has a lot to work with in improving their obstacle course to be more equitable to all Cadets.

In short, the ROTC should look more into the obstacle course they use to highlight key areas where shorter Cadets are struggling at. Then modify those problematic areas in a way to not penalize the shorter Cadets and ensure that part is not too easy for the other Cadets to pass through.

Appendix

R Code 01

```
#Getting the data into the document  
APFT <- read_csv(file.path(str_remove(getwd(), "/Class Notes"), "Book Data/APFT.csv"))
```

R Code 02

```
#Defining certain numerical variables into categorical. This pertains to the CS variable.  
APFT <- APFT %>%  
  mutate(CS = as.factor(CS))
```

Figure 01

```
#Generation of the scatter plot between height and course completion time  
APFT %>%  
  ggplot(aes(x=height, y=IOCT_Time))+  
  geom_point()+  
  labs(  
    title="Scatter-plot between height and IOCT time",  
    x="Heights of Cadets in inches",  
    y="Seconds needed to complete the obstacle course"  
  )+  
  theme_minimal()
```

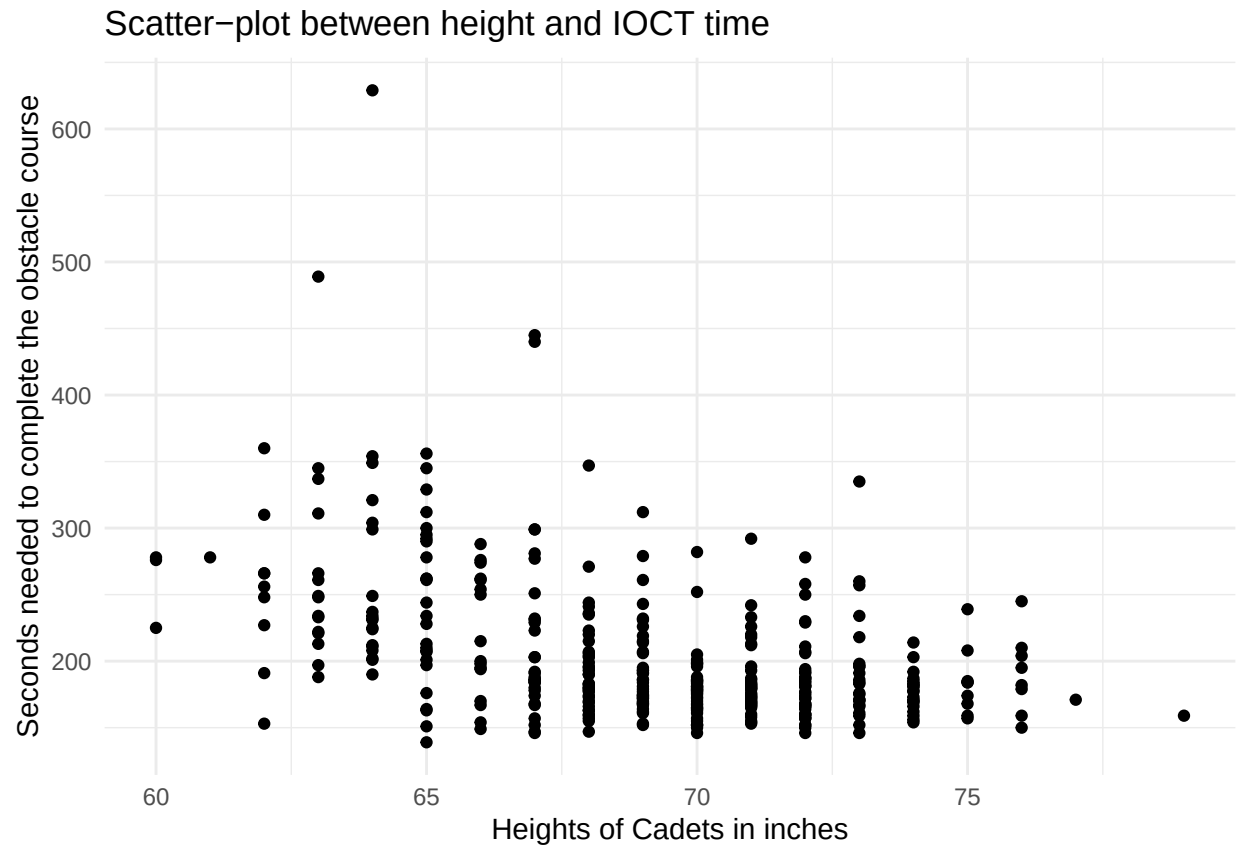


Figure 02

```
#Generation of the scatter plot between height and weight
APFT %>%
  ggplot(aes(x=height, y=weight))+
  geom_point()+
  labs(
    title="Scatter-plot between heights and weights of Cadets",
    x="Heights of Cadets in inches",
    y="Weights of Cadets in pounds"
  )+
  theme_minimal()
```

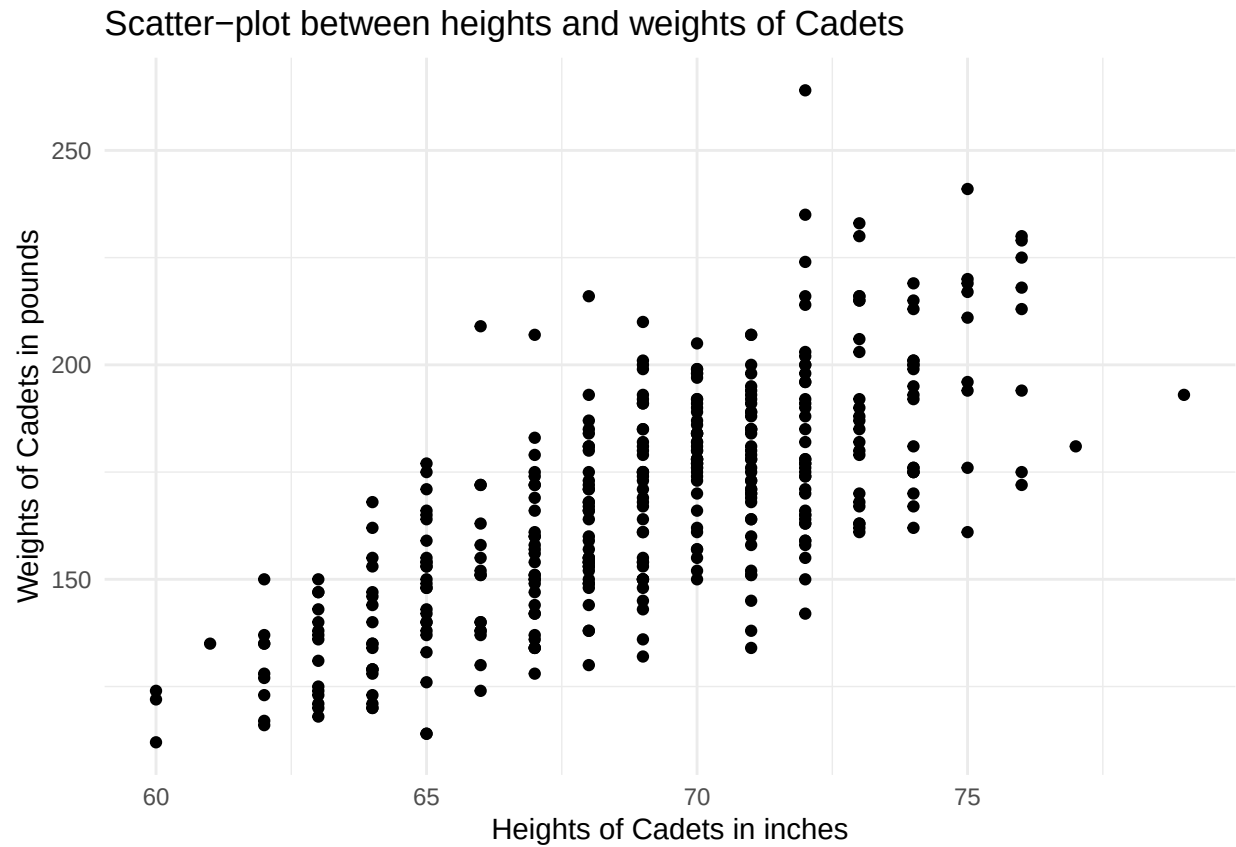


Figure 03

```
#Generation of the scatter plot between weight and course completion time
APFT %>%
  ggplot(aes(x=weight, y=IOCT_Time))+
  geom_point()+
  labs(
    title="Scatter-plot between weight and IOCT Time",
    x="Weights of Cadets in pounds",
    y="Seconds needed to complete the obstacle course"
  )+
  theme_minimal()
```

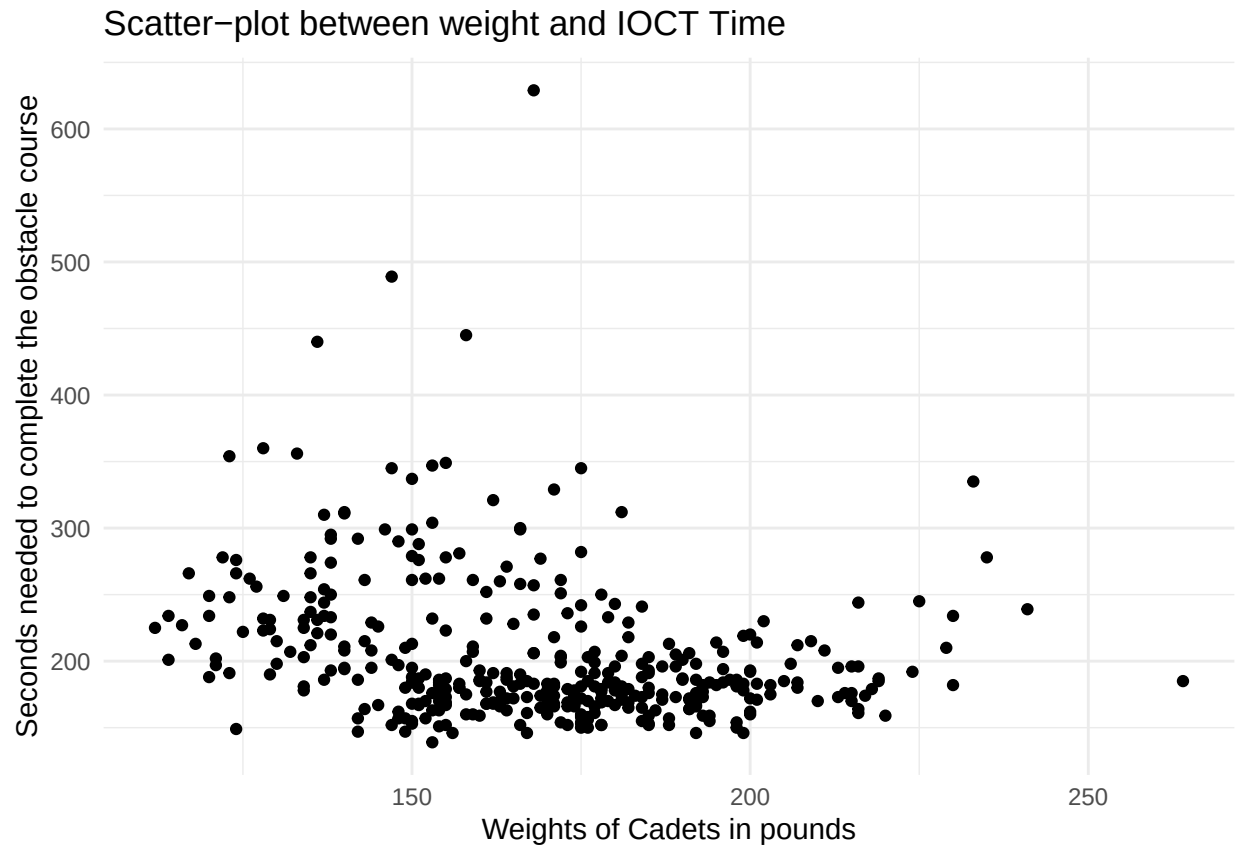
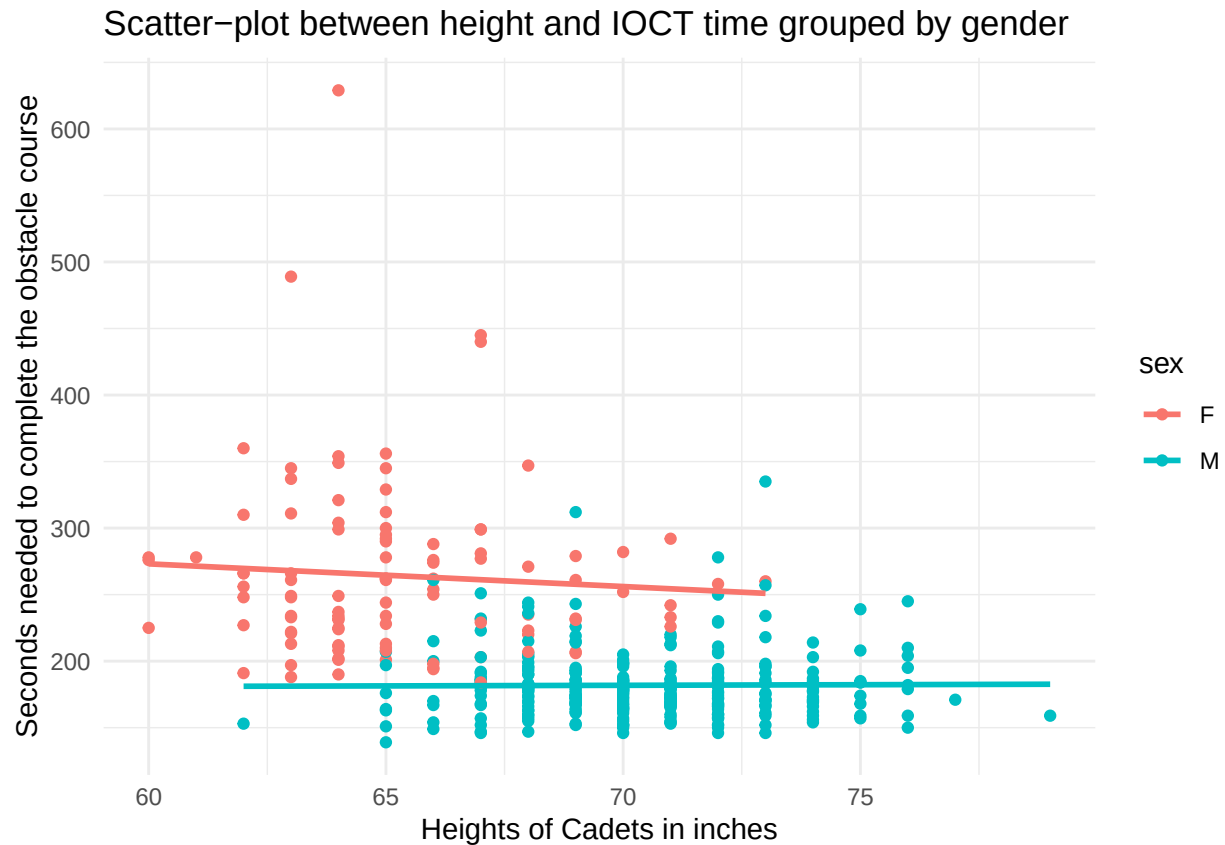


Figure 04

```
#Generation of the scatter plot between height and course completion time grouped by gender
APFT %>%
  group_by(sex) %>%
  ggplot(aes(x=height, y=IOCT_Time, col=sex))+
  geom_point()+
  labs(
    title="Scatter-plot between height and IOCT time grouped by gender",
    x="Heights of Cadets in inches",
    y="Seconds needed to complete the obstacle course"
  )+
  theme_minimal()+
  geom_smooth(method="lm", se=FALSE)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



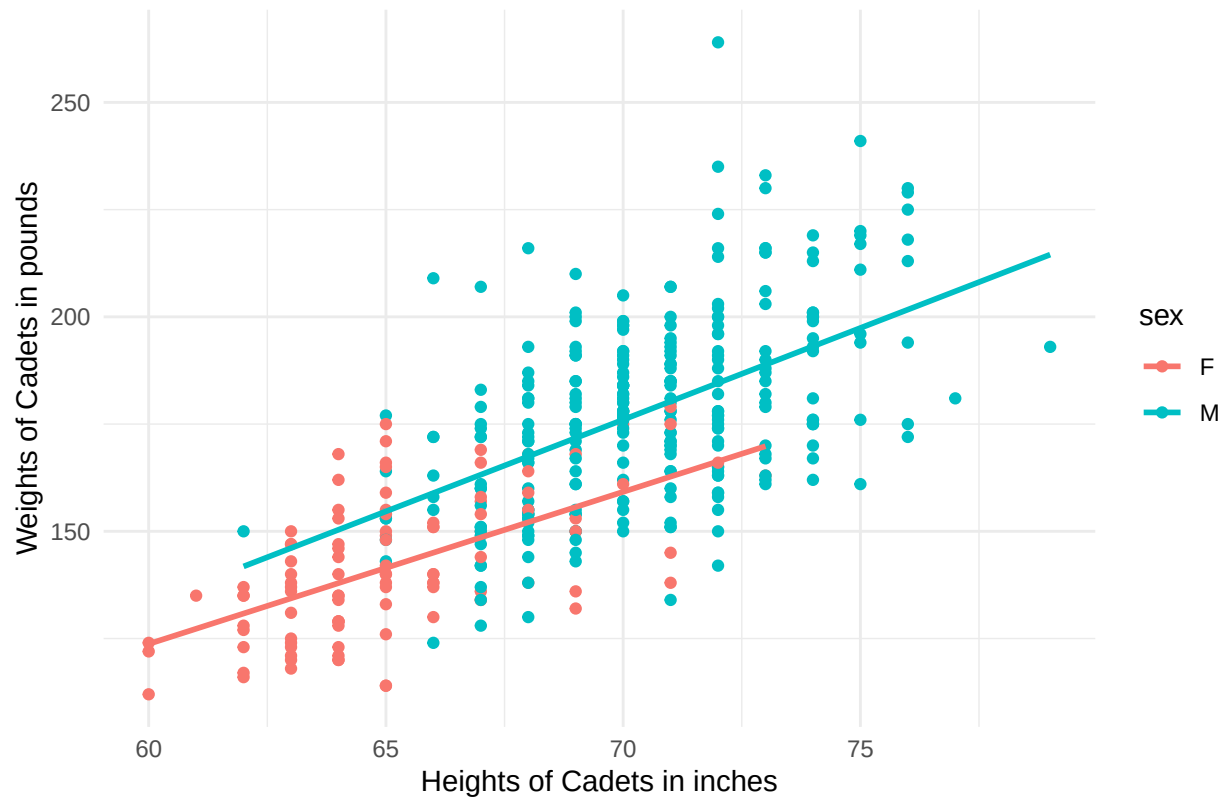
Looking at *Figure 04*, we see that there is a difference of slopes between height of Cadets and obstacle course completion times based on their gender.

Figure 05

```
#Generation of the scatter plot between height and weight grouped by gender
APFT %>%
  group_by(sex) %>%
  ggplot(aes(x=height, y=weight, col=sex))+
  geom_point()+
  labs(
    title="Scatter-plot between heights and weights of Cadets grouped by gender",
    x="Heights of Cadets in inches",
    y="Weights of Cadets in pounds"
  )+
  theme_minimal()+
  geom_smooth(method="lm", se=FALSE)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Scatter-plot between heights and weights of Cadets grouped by gender

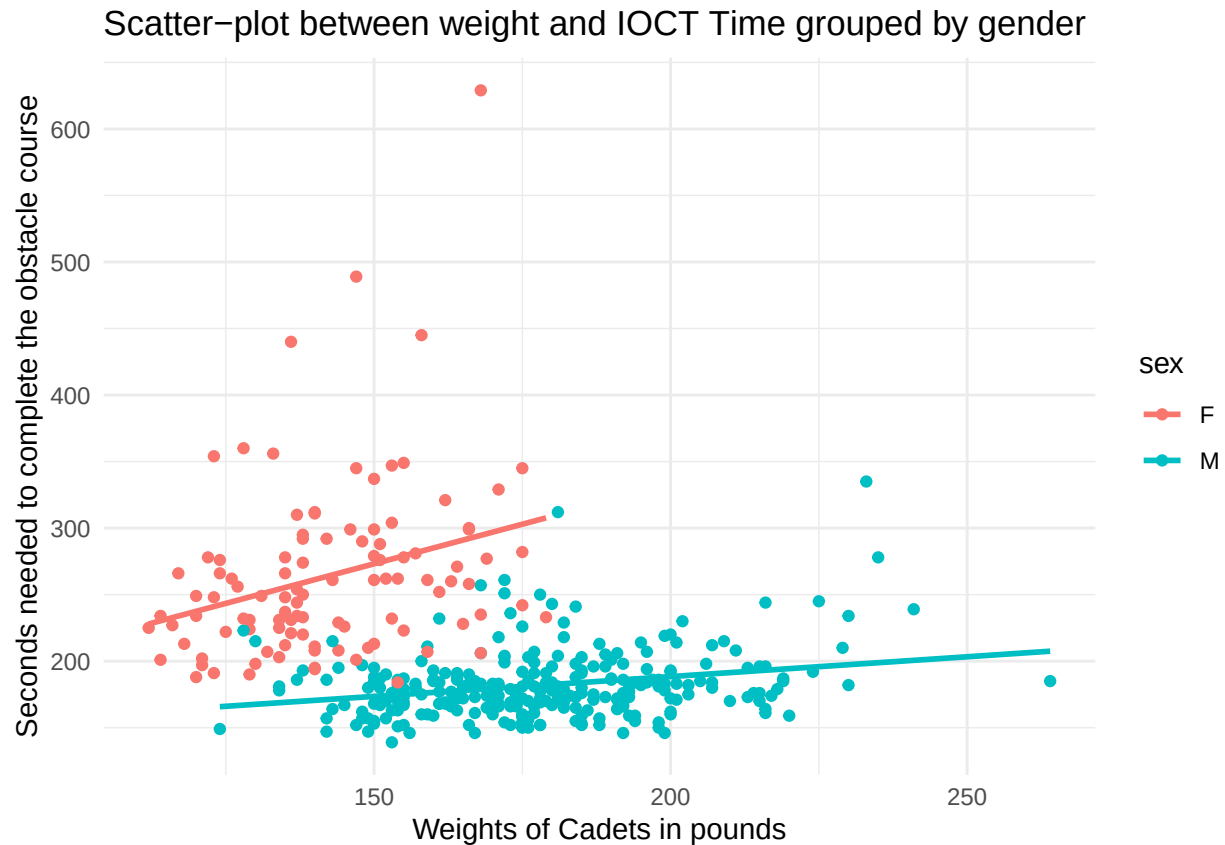


From *Figure 05*, we see that there is not much of a difference between the height and weights of the Cadets based on their Gender.

Figure 06

```
#Generation of the scatter plot between weight and course completion time grouped by gender
APFT %>%
  group_by(sex) %>%
  ggplot(aes(x=weight, y=IOCT_Time, col=sex))+
  geom_point()+
  labs(
    title="Scatter-plot between weight and IOCT Time grouped by gender",
    x="Weights of Cadets in pounds",
    y="Seconds needed to complete the obstacle course"
  )+
  theme_minimal()+
  geom_smooth(method="lm", se=FALSE)

## 'geom_smooth()' using formula = 'y ~ x'
```



From *Figure 06*, we see a difference of slopes between the weights of Cadets and their obstacle course completion time based on their gender.

R Code 03

```
#Creating a full model
APFTReg1 <- lm(IOCT_Time ~ ., data=APFT)
summary(APFTReg1)
```

```
##
## Call:
## lm(formula = IOCT_Time ~ ., data = APFT)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -68.22  -18.90   -4.42   12.47  343.44
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  471.1184    52.9609   8.896 < 2e-16 ***
## sexM         -94.5053     5.8761 -16.083 < 2e-16 ***
## height       -2.6726     0.8680  -3.079  0.00223 **
## weight        0.5732     0.1102   5.203 3.23e-07 ***
## pu_score     -0.3765     0.1848  -2.038  0.04227 *
## su_score     -0.2409     0.1624  -1.484  0.13866
```

```
## run_score      -0.5742      0.1442   -3.983 8.17e-05 ***
## CS1            -8.4099      4.6060   -1.826 0.06867 .
## APFT_Score      NA          NA       NA      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 36.49 on 376 degrees of freedom
## Multiple R-squared:  0.5647, Adjusted R-squared:  0.5566
## F-statistic: 69.67 on 7 and 376 DF,  p-value: < 2.2e-16
```

R Code 03 formulaic

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + \beta_5 x_{5i} + \beta_6 x_{6i} + \beta_7 x_{7i} + \beta_8 x_{8i} + \epsilon_i$$

y_i = Obstacle course time for ith Cadet
 x_{1i} = ith Cadet is a male
 x_{2i} = height of ith Cadet
 x_{3i} = weight of ith Cadet
 x_{4i} = score for push ups completed in 2 minutes of ith Cadet
 x_{5i} = score for sit ups completed in 2 minutes of ith Cadet
 x_{6i} = score for ith Cadet time to run two miles
 x_{7i} = ith Cadet was a recruited athlete after highschool
 x_{8i} = ith Cadet combined scores for push ups, sit ups, and running
 $\epsilon_i \sim N(0, \sigma^2)$

The reason as to why the APFT score coefficients show up as NA, is because $APFT_i = puscore_i + suscore_i + runscore_i$

R Code 04

```
#Full model without APFT_Score
APFTReg2 <- lm(IOCT_Time ~ .-APFT_Score, data=APFT)
summary(APFTReg2)

##
## Call:
## lm(formula = IOCT_Time ~ . - APFT_Score, data = APFT)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -68.22 -18.90  -4.42  12.47 343.44
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  471.1184    52.9609   8.896 < 2e-16 ***
## sexM         -94.5053     5.8761 -16.083 < 2e-16 ***
## height       -2.6726     0.8680  -3.079 0.00223 **
## weight         0.5732     0.1102   5.203 3.23e-07 ***
```

```
## pu_score      -0.3765      0.1848  -2.038  0.04227 *
## su_score      -0.2409      0.1624  -1.484  0.13866
## run_score     -0.5742      0.1442  -3.983  8.17e-05 ***
## CS1           -8.4099      4.6060  -1.826  0.06867 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 36.49 on 376 degrees of freedom
## Multiple R-squared:  0.5647, Adjusted R-squared:  0.5566
## F-statistic: 69.67 on 7 and 376 DF,  p-value: < 2.2e-16
```

R Code 04 formulaic

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + \beta_5 x_{5i} + \beta_6 x_{6i} + \beta_7 x_{7i} + \epsilon_i$$

y_i = Obstacle course time for ith Cadet
 x_{1i} = ith Cadet is a male
 x_{2i} = height of ith Cadet
 x_{3i} = weight of ith Cadet
 x_{4i} = score for push ups completed in 2 minutes of ith Cadet
 x_{5i} = score for sit ups completed in 2 minutes of ith Cadet
 x_{6i} = score for ith Cadet time to run two miles
 x_{7i} = ith Cadet was a recruited athlete after highschool
 $\epsilon_i \sim N(0, \sigma^2)$

This model is still too large to do any statistical interpretations as the model is over-fit to explain a Cadet's completion time of the obstacle course. However, looking at *figures 04, 05, and 06* gives us significant evidence of what our final model should be.

R Code 05

```
#Generating the final model with interaction terms between height-sex, weight-sex, and height-weight
APFTRegF <- lm(IOCT_Time~(height+weight+sex)^2, data=APFT)
summary(APFTRegF)
```

```
##
## Call:
## lm(formula = IOCT_Time ~ (height + weight + sex)^2, data = APFT)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -88.86 -18.23  -4.49   11.79  298.51
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   255.19153   326.18185    0.782  0.434494
## height        -4.38427     5.09577   -0.860  0.390129
```

```
## weight      4.17050      2.16074      1.930 0.054341 .
## sexM        -432.03822    149.87453    -2.883 0.004169 **
## height:weight -0.03206      0.03281    -0.977 0.329057
## height:sexM    8.44755      2.23055      3.787 0.000177 ***
## weight:sexM    -1.49889      0.36114     -4.150 4.1e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 37.9 on 377 degrees of freedom
## Multiple R-squared:  0.529, Adjusted R-squared:  0.5215
## F-statistic: 70.58 on 6 and 377 DF, p-value: < 2.2e-16
```

```
anova(APFTRegF, APFTReg2)
```

```
## Analysis of Variance Table
##
## Model 1: IOCT_Time ~ (height + weight + sex)^2
## Model 2: IOCT_Time ~ (sex + height + weight + pu_score + su_score + run_score +
##      CS + APFT_Score) - APFT_Score
##      Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1      377 541552
## 2      376 500573   1    40979 30.781 5.459e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

R Code 05 formulaic

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{1i} x_{2i} + \beta_5 x_{1i} x_{3i} + \beta_6 x_{2i} x_{3i} + \epsilon_i$$

y_i = Obstacle course time for ith Cadet

x_{1i} = height of ith Cadet

x_{2i} = weight of ith Cadet

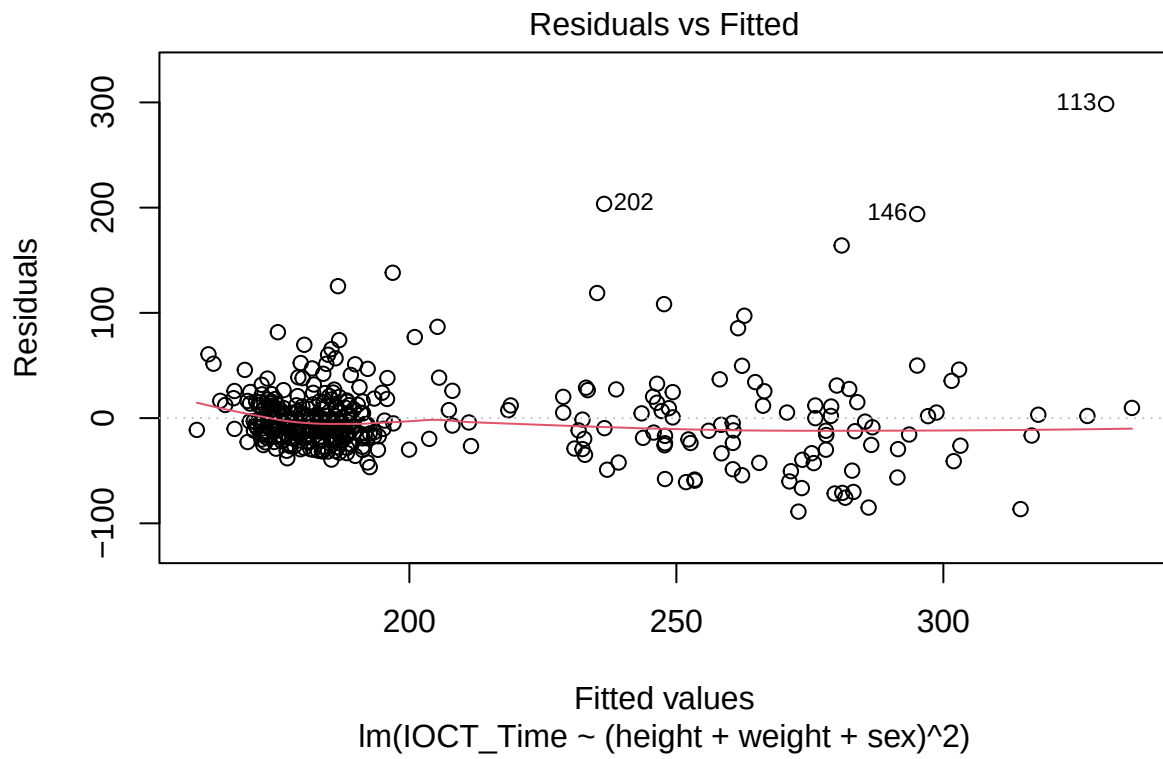
x_{3i} = ith Cadet is a male

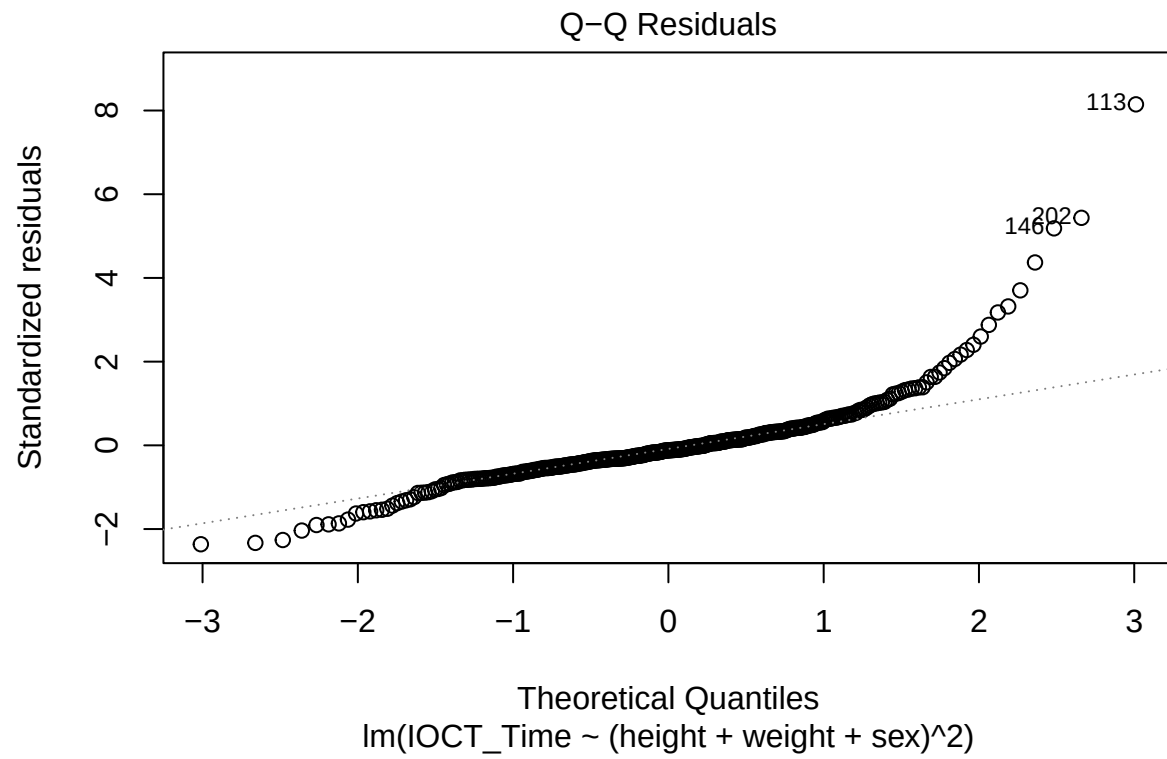
$$\epsilon_i \sim N(0, \sigma^2)$$

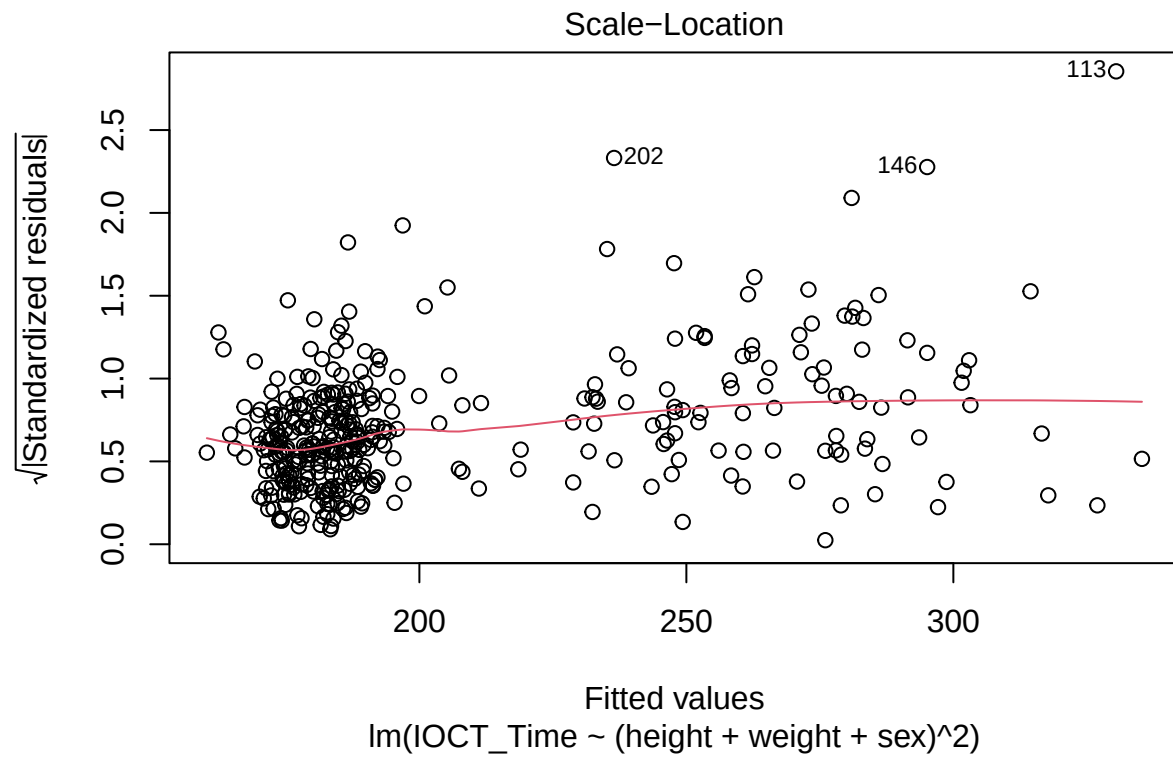
From the one-way analysis of variance, we see that the new model does a better job at explaining the variation of the model than the old full model. This means that the new interaction model does a better job at fitting the data.

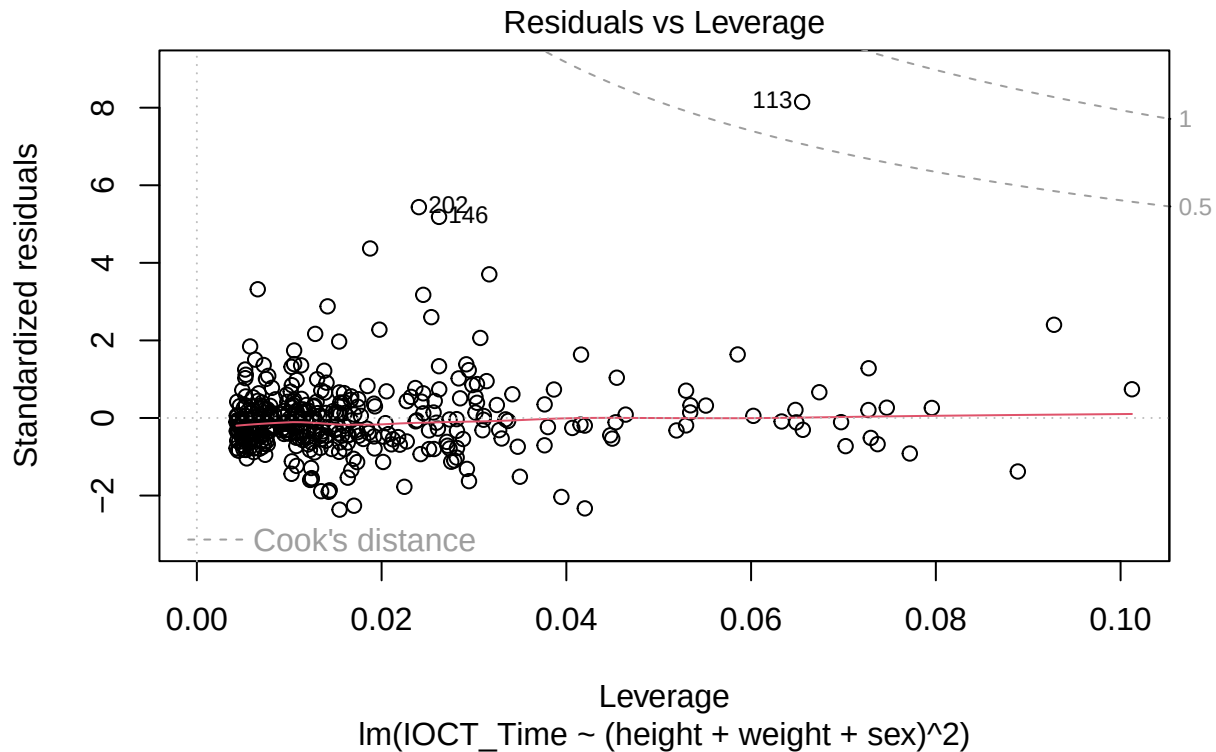
Figures 07-10

```
plot(APFTRegF)
```









From the above figures, we see that our model does have one outliers. Specifically at data point 113 via Cook's Distance.

R Code 06

```
#Removing the outlier at the 113th row and regenerating the final model
APFT$IOCT_Time[113]
```

```
## [1] 629
```

```
APFT2 <- APFT %>%
  filter(IOCT_Time<600)
APFTRegF2 <- lm(IOCT_Time~(height+weight+sex)^2, data=APFT2)
summary(APFTRegF2)
```

```
##
## Call:
## lm(formula = IOCT_Time ~ (height + weight + sex)^2, data = APFT2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -83.608 -18.160  -4.415  12.550 205.483
##
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   288.61713   296.49163    0.973  0.33096
## height        -3.75198     4.63210   -0.810  0.41846
## weight         2.79182     1.96992    1.417  0.15724
## sexM          -306.15422   136.94357   -2.236  0.02596 *
## height:weight  -0.01934     0.02985   -0.648  0.51747
## height:sexM     5.56345     2.05273    2.710  0.00703 **
## weight:sexM    -1.02246     0.33251   -3.075  0.00226 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 34.45 on 376 degrees of freedom
## Multiple R-squared:  0.5395, Adjusted R-squared:  0.5321
## F-statistic: 73.41 on 6 and 376 DF,  p-value: < 2.2e-16
```

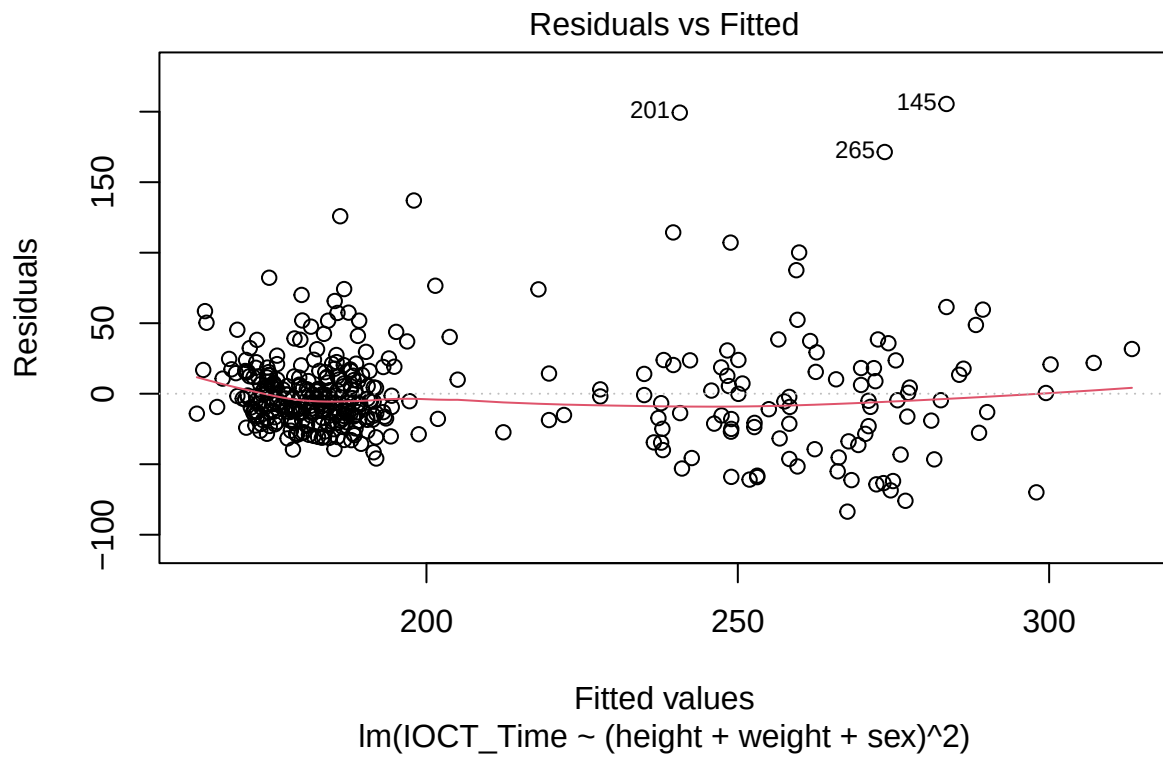
R Code 06 formulaic

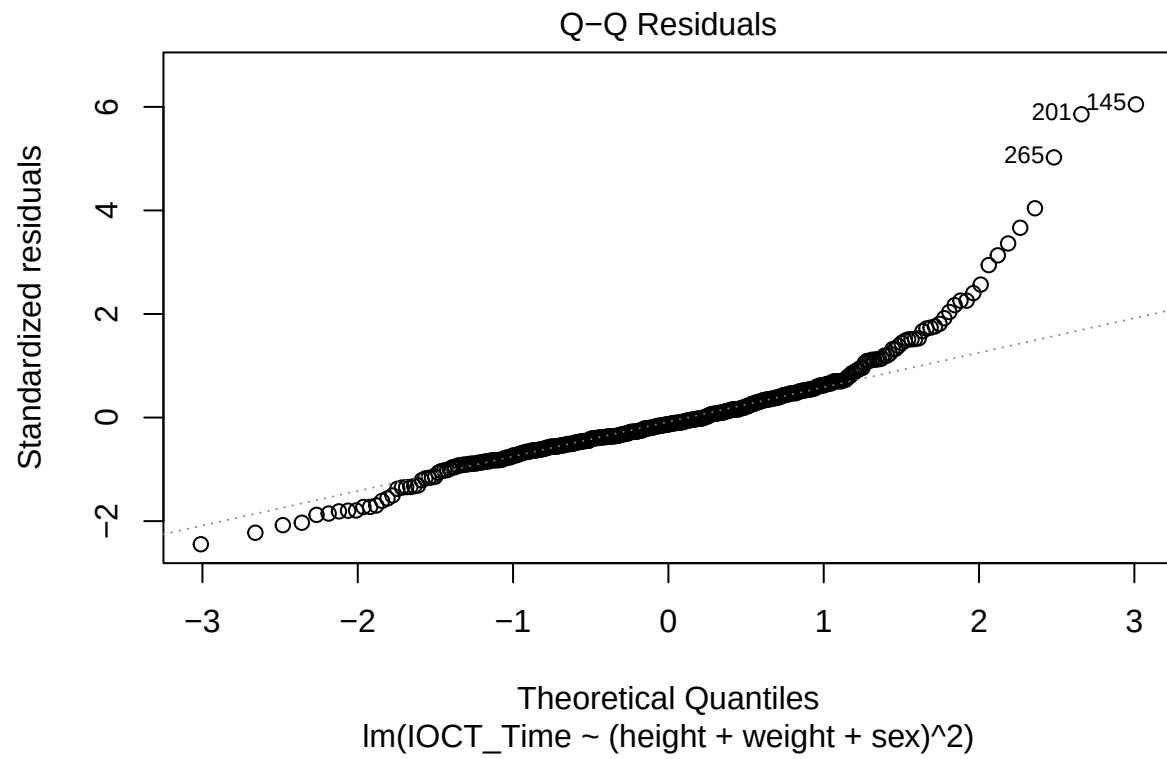
$$\widehat{y_{Femalei}} = 288.6171 - 3.7520x_{Heighti} + 2.7918x_{Weighti} - 0.0193x_{Heighti} * x_{Weighti}$$

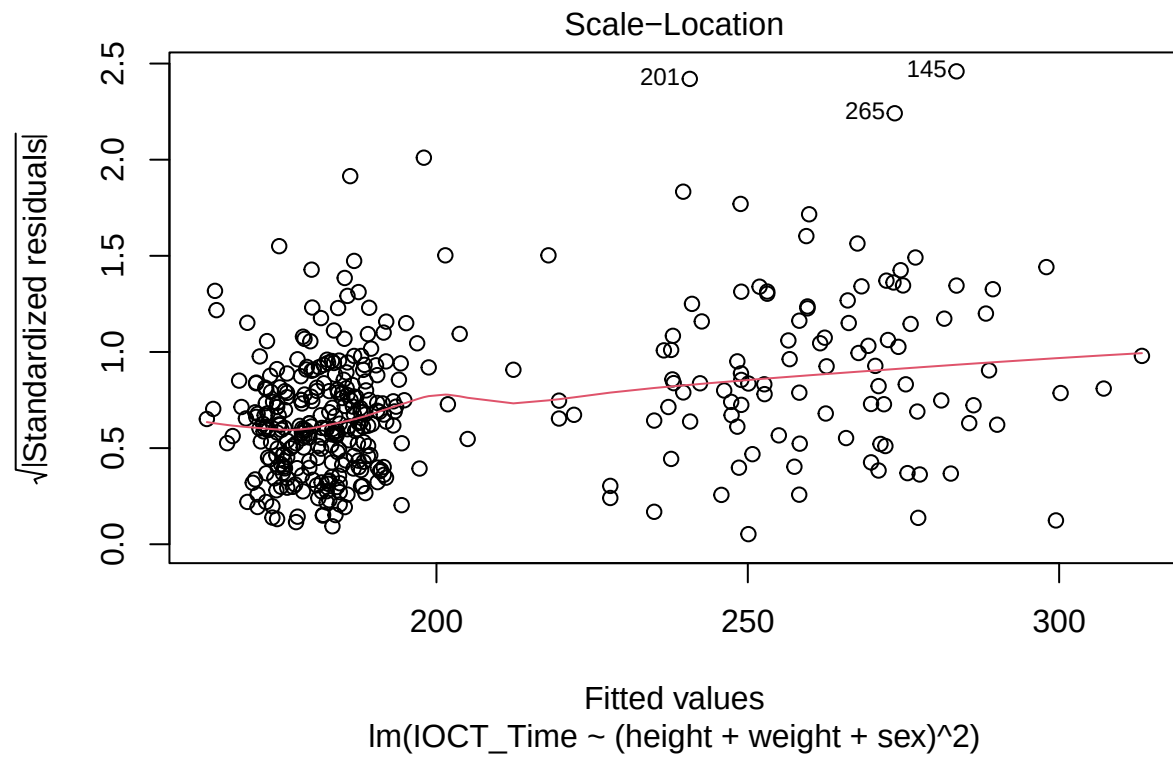
$$\widehat{y_{Malei}} = -17.5371 + 1.8825x_{Heighti} + 1.7693x_{Weighti} - 0.0193x_{Heighti} * x_{Weighti}$$

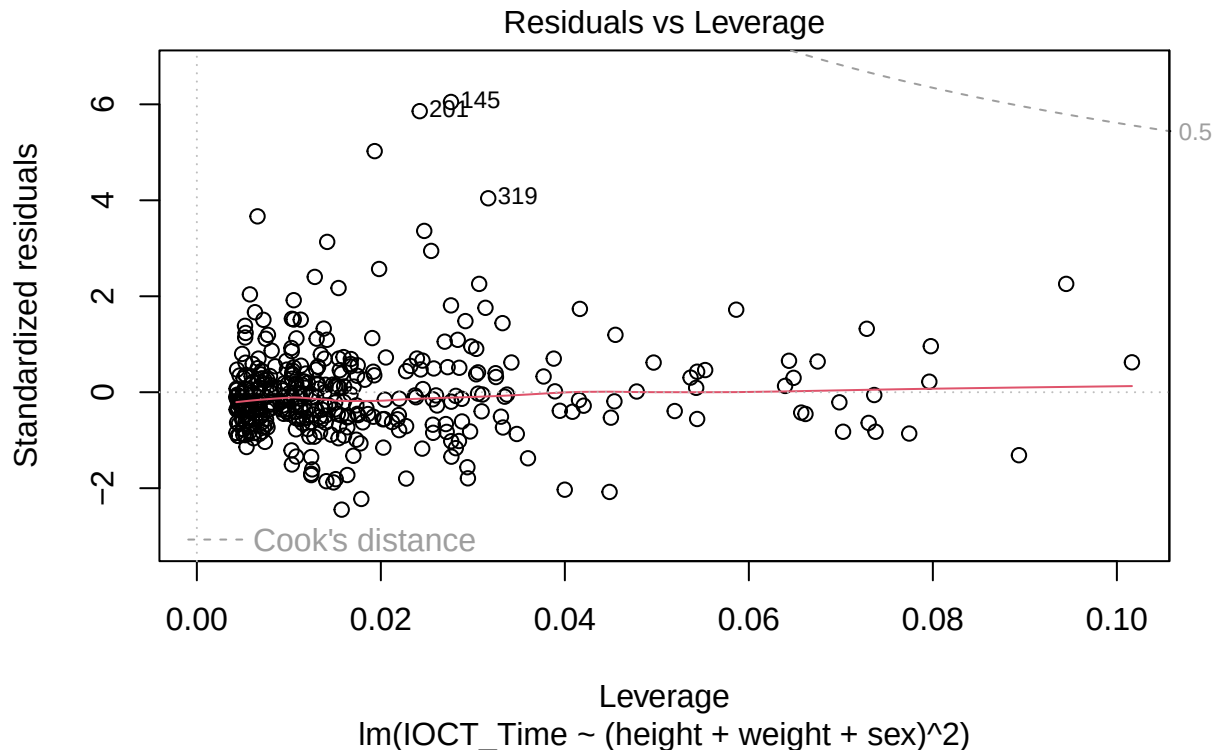
Figures 11-14

```
plot(APFTRegF2)
```









Looking at *Figure 11-14*, we see that removing the outlier led to a better fit overall. The Q-Q plot on *Figure 12* right tail shrunk down more than *Figure 08*. While *Figures 07 & 11* have some fanning out, it is not unexpected. We are comparing between two distinct groups (Male/Female) which have different variances that can be shown from *Figures 04-06*.

R Code 07

Point estimates of obstacle course times for both Male and Female Cadets weighing 160 pounds and are 68 inches high are:

```
#Computing the point estimates in course completion times between similar male and female Cadets
PEMale <- c(1, 68, 160, 1, 160*68, 68, 160)
PEFemale <- c(1, 68, 160, 0, 160*68, 0, 0)

t(PEMale) %*% coef(APFTRegF2)
```

```
##           [,1]
## [1,] 178.3036
```

```
t(PEFemale) %*% coef(APFTRegF2)
```

```
##           [,1]
## [1,] 269.7371
```

We can predict that a male Cadet that is 68 inches tall and weighs 160 pounds to complete the course in 178.30 seconds while the similar female Cadet completes the course in 269.74 seconds.

R Code 08

Confidence intervals for both male and female Cadets weighing 160 pounds and are 68 inches tall are:

```
#Generating the confidence intervals in course completion times between similar male and female Cadets
APFTRegF2Cov <- vcov(APFTRegF2)
seMale <- sqrt(t(PEMale)%*%APFTRegF2Cov%*%PEMale)
seFemale <- sqrt(t(PEFemale)%*%APFTRegF2Cov%*%PEFemale)

t(PEMale) %*% coef(APFTRegF2) + c(-1, 1)*qt(0.975, df.residual(APFTRegF2))*seMale
```

```
## [1] 172.4285 184.1787
```

```
t(PEFemale) %*% coef(APFTRegF2) + c(-1, 1)*qt(0.975, df.residual(APFTRegF2))*seFemale
```

```
## [1] 259.3091 280.1652
```

We are 95% confident that a male Cadet that is 68 inches tall and weighs 160 pounds can complete the obstacle course between 172.43 seconds to 184.17 seconds.

We are 95% confident that a female Cadet that is 68 inches tall and weighs 160 pounds can complete the obstacle course between 259.31 seconds to 280.17 seconds.

R Code 09

```
#Looking at the range of heights between male and female Cadets
APFT2 %>%
  group_by(sex) %>%
  summarize(Minimum = min(height), Maximum = max(height))
```

```
## # A tibble: 2 x 3
##   sex    Minimum Maximum
##   <chr>   <dbl>   <dbl>
## 1 F         60      73
## 2 M         62      79
```

R Code 10

```
#Looking at the difference in effect of height for similar male and female Cadets

PEMale2 <- c(1, 63, 160, 1, 63*160, 63, 160) - PEMale#Male Cadets 5 inches shorter
PEFemale2 <- c(1, 63, 160, 0, 63*160, 0, 0) - PEFemale#Female Cadets 5 inches shorter

#Point estimates for Cadets 5 inches shorter
t(PEMale2) %*% coef(APFTRegF2)
```

```
##           [,1]
## [1,] 6.415961
```

```
t(PEFemale2) %*% coef(APFTRegF2)
```

```
##           [,1]  
## [1,] 34.23322
```

```
#Confidence intervals for Cadets 5 inches shorter
```

```
t(PEMale2) %*% coef(APFTRegF2) + c(-1, 1)*qt(0.975, df.residual(APFTRegF2))*seMale
```

```
## [1] 0.5408271 12.2910952
```

```
t(PEFemale2) %*% coef(APFTRegF2) + c(-1, 1)*qt(0.975, df.residual(APFTRegF2))*seFemale
```

```
## [1] 23.80517 44.66128
```

R Code 10 Interpretations:

We can expect a Male Cadet that is 63 inches tall and weighs 160 pounds to complete the obstacle course 6.42 seconds slower than the Male Cadet that is 68 inches tall and weighs 160 pounds.

We can expect a Female Cadet that is 63 inches tall and weighs 160 pounds to complete the obstacle course 34.23 seconds slower than the Female Cadet that is 68 inches tall and weighs 160 pounds.

We are 95% confident that a Male Cadet that is 63 inches tall and weighs 160 pounds to complete the obstacle course between 0.54 to 12.29 seconds slower than the Male Cadet who is 68 inches tall and weighs 160 pounds.

We are 95% confident that a Female Cadet that is 63 inches tall and weighs 160 pounds to complete the obstacle course between 23.81 to 44.66 seconds slower than the Female Cadet who is 68 inches tall and weighs 160 pounds.